

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

SOLVING TRIGONOMETRIC EQUATION TYPE 1 (FACTORISATION)

1. Find the general values of θ for which the quadratic function $(\sin\theta)x^2 + (2\cos\theta)x + \frac{\cos\theta + \sin\theta}{2}$ is the square of a linear function.

θ के वह व्यापक मान ज्ञात कीजिए जिनके लिए द्विघात फलन $(\sin\theta)x^2 + (2\cos\theta)x + \frac{\cos\theta + \sin\theta}{2}$ एक रेखीय फलन का वर्ग है।

Ans. $2n\pi + \frac{\pi}{4}$ or $(2n+1)\pi - \tan^{-1}2$, $n \in I$

Sol. Given quadratic function

$(\sin\theta)x^2 + (2\cos\theta)x + \frac{\cos\theta + \sin\theta}{2}$ is the square of a linear function.

$$\Rightarrow \sin\theta > 0 \text{ \& } D \equiv 4\cos^2\theta - 4\sin\theta \left(\frac{\sin\theta + \cos\theta}{2} \right) = 0$$

$$\Rightarrow \sin\theta > 0 \text{ \& } 4\cos^2\theta - 2\sin^2\theta - 2\sin\theta\cos\theta = 0$$

$$\Rightarrow \sin\theta > 0 \text{ \& } (4\cos\theta + 2\sin\theta)(\cos\theta - \sin\theta) = 0$$

$$\Rightarrow \sin\theta > 0 \text{ \& } (\cos\theta - \sin\theta)(\sin\theta + 2\cos\theta) = 0$$

$$\Rightarrow \sin\theta > 0 \text{ \& } \tan\theta = 1 \text{ or } \tan\theta = -2$$

$$\Rightarrow \sin\theta > 0 \text{ \& } \tan\theta = 1$$

or

$$\sin\theta > 0 \text{ \& } \tan\theta = -2$$

$$\Rightarrow \theta = 2n\pi + \frac{\pi}{4}; n \in I$$

or

$$\theta = (2n+1)\pi - \tan^{-1}(2); n \in I$$

2. Solve the equality: $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$.

समीकरण : $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$ को हल कीजिए।

Ans. $x = \frac{n\pi}{7} - \frac{\pi}{84}$ or $x = \frac{n\pi}{4} + \frac{7\pi}{48}$, $n \in I$

Sol. $2\sin 11x + \cos 3x + \sqrt{3}\sin 3x = 0$

$$\Rightarrow 2\sin 11x + 2\left(\frac{1}{2}\cos 3x + \frac{\sqrt{3}}{2}\sin 3x\right) = 0$$

$$\Rightarrow 2\sin 11x + 2\sin\left(3x + \frac{\pi}{6}\right) = 0$$

$$\Rightarrow \sin 11x + \sin\left(3x + \frac{\pi}{6}\right) = 0$$

$$\Rightarrow 2 \sin\left(\frac{11x + 3x + \frac{\pi}{6}}{2}\right) \cos\left(\frac{11x - 3x - \frac{\pi}{6}}{2}\right) = 0$$

$$\Rightarrow \sin\left(\frac{14x + \frac{\pi}{6}}{2}\right) = 0 \quad \text{or} \quad \cos\left(\frac{8x - \frac{\pi}{6}}{2}\right) = 0$$

$$\Rightarrow \frac{14x + \frac{\pi}{6}}{2} = n\pi; n \in I \quad \text{or} \quad \frac{8x - \frac{\pi}{6}}{2} = (2n+1)\frac{\pi}{2}; n \in I$$

$$\Rightarrow x = \frac{n\pi}{7} - \frac{\pi}{84}; n \in I \quad \text{or} \quad x = \frac{n\pi}{4} + \frac{7\pi}{48}; n \in I$$

SOLVING TRIGONOMETRIC EQUATION TYPE 3 (SUM TO PRODUCT AND PRODUCT TO SUM)

3. Find all the values of θ satisfying the equation; $\sin \theta + \sin 5\theta = \sin 3\theta$ such that $0 \leq \theta \leq \pi$.

θ के सभी मान ज्ञात कीजिए जो समीकरण $\sin \theta + \sin 5\theta = \sin 3\theta$ को संतुष्ट करते हैं, जहाँ $0 \leq \theta \leq \pi$.

Ans. $0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}$ and π

Sol. $\sin \theta + \sin 5\theta = \sin 3\theta; 0 \leq \theta \leq \pi$
 $2 \sin 3\theta \cos 2\theta - \sin 3\theta = 0$

$$\Rightarrow \sin 3\theta = 0 \quad \text{or} \quad \cos 2\theta = \frac{1}{2}$$

Given $0 \leq \theta \leq \pi$

$$\Rightarrow 3\theta = 0, \pi, 2\pi, 3\pi \quad \text{or} \quad 2\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\Rightarrow \theta = 0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{6}, \pi$$

4. Find all values of θ between 0° and 180° satisfying the equation; $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$.

अन्तराल 0° तथा 180° में समीकरण $\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$ को संतुष्ट करने वाले θ के सभी मान ज्ञात कीजिए।

Ans. $30^\circ, 45^\circ, 90^\circ, 135^\circ, 150^\circ$

Sol. Given

$$\theta \in [0, \pi]$$

$$\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$$

$$\Rightarrow 2\cos 5\theta \cdot \cos \theta + 2\cos^2 \theta = 0$$

$$\Rightarrow 2\cos \theta (\cos 5\theta + \cos \theta) = 0$$

$$\Rightarrow 2\cos \theta \cdot 2\cos 3\theta \cos 2\theta = 0$$

$$\Rightarrow \cos \theta \cdot \cos 2\theta \cdot \cos 3\theta = 0$$

$$\Rightarrow \cos \theta = 0 \quad \text{or} \quad \cos 2\theta = 0 \quad \text{or} \quad \cos 3\theta = 0$$

$$\Rightarrow \theta = 90^\circ \quad \text{or} \quad 2\theta = 90^\circ, 270^\circ \quad \text{or} \quad 3\theta = 90^\circ, 270^\circ, 450^\circ$$

$$\Rightarrow \theta = 30^\circ, 45^\circ, 90^\circ, 135^\circ, 150^\circ$$

5. The positive integer value of $n > 3$ satisfying the equation $\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$ is.

समीकरण $\frac{1}{\sin\left(\frac{\pi}{n}\right)} = \frac{1}{\sin\left(\frac{2\pi}{n}\right)} + \frac{1}{\sin\left(\frac{3\pi}{n}\right)}$ को सन्तुष्ट करने वाले धनात्मक पूर्णांक $n(>3)$ का मान होगा।

Ans. 7

Sol. Given $\frac{1}{\sin \frac{\pi}{n}} = \frac{1}{\sin \frac{2\pi}{n}} + \frac{1}{\sin \frac{3\pi}{n}}$ & $n > 3$

$$\Rightarrow \frac{1}{\sin \frac{\pi}{n}} - \frac{1}{\sin \frac{3\pi}{n}} = \frac{1}{\sin \frac{2\pi}{n}}$$

$$\Rightarrow \frac{\sin \frac{3\pi}{n} - \sin \frac{\pi}{n}}{\sin \frac{\pi}{n} \sin \frac{3\pi}{n}} = \frac{1}{\sin \frac{2\pi}{n}}$$

$$\Rightarrow \frac{2 \cos \frac{2\pi}{n} \sin \frac{\pi}{n}}{\sin \frac{\pi}{n} \sin \frac{3\pi}{n}} = \frac{1}{\sin \frac{2\pi}{n}}$$

$$\Rightarrow 2 \cos \frac{2\pi}{n} \sin \frac{2\pi}{n} = \sin \frac{3\pi}{n}$$

$$\Rightarrow \sin \frac{4\pi}{n} = \sin \frac{3\pi}{n} \Rightarrow \frac{4\pi}{n} = K\pi + (-1)^K \frac{3\pi}{n}$$

$$\text{If } K = 2m \quad \text{then } \frac{\pi}{n} = 2m\pi$$

$$\Rightarrow n = \frac{1}{2m} \quad \Rightarrow n = \frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$$

$$\text{If } K = 2m + 1 \quad \text{then } \frac{7\pi}{n} = (2m + 1)\pi$$

$$\Rightarrow n = \frac{7}{2m + 1} \quad \Rightarrow n = 7, \frac{7}{3}, \frac{7}{5}, \dots$$

$$\text{Given } n > 3 \Rightarrow n = 7$$

SOLVING TRIGONOMETRIC EQUATION TYPE 4 (SUBSTITUTION BASED)

6. Find all value of θ , between 0 & π , which satisfy the equation; $\cos \theta \cdot \cos 2\theta \cdot \cos 3\theta = 1/4$.

अन्तराल 0 व π के मध्य θ के सभी मान ज्ञात कीजिए जो समीकरण $\cos \theta \cdot \cos 2\theta \cdot \cos 3\theta = 1/4$ को संतुष्ट करते हैं।

Ans. $\frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$

Sol. $\cos \theta \cdot \cos 2\theta \cdot \cos 3\theta = \frac{1}{4}$

$$\cos \theta (2\cos^2\theta - 1) \cdot (4\cos^3\theta - 3\cos\theta) = \frac{1}{4}$$

$$\Rightarrow (2\cos^3\theta - \cos\theta) (4\cos^3\theta - 3\cos\theta) = \frac{1}{4}$$

$$\Rightarrow 8\cos^6\theta - 6\cos^4\theta - 4\cos^4\theta + 3\cos^2\theta = \frac{1}{4}$$

$$\Rightarrow 8\cos^6\theta - 10\cos^4\theta + 3\cos^2\theta = \frac{1}{4}$$

$$\text{Let } \cos^2\theta = t$$

$$\Rightarrow 8t^3 - 10t^2 + 3t = \frac{1}{4}$$

$$\Rightarrow 32t^3 - 40t^2 + 12t - 1 = 0$$

$$\Rightarrow \left(t - \frac{1}{4}\right) (32t^2 - 32t + 4) = 0$$

$$\Rightarrow \left(t - \frac{1}{4}\right) (8t^2 - 8t + 1) = 0$$

$$\Rightarrow t = \frac{1}{4} \quad \text{or} \quad t = \frac{\sqrt{2}+1}{2\sqrt{2}} \quad \text{or} \quad t = \frac{\sqrt{2}-1}{2\sqrt{2}}$$

$$\Rightarrow \cos^2 \theta = \frac{1}{4} \quad \text{or} \quad \cos^2 \theta = \frac{\sqrt{2}+1}{2\sqrt{2}} \quad \text{or} \quad \cos^2 \theta = \frac{\sqrt{2}-1}{2\sqrt{2}}$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{3} \quad \text{or} \quad \theta = n\pi \pm \frac{\pi}{8} \quad \text{or} \quad \theta = n\pi \pm \frac{3\pi}{8}$$

Given $\theta \in (0, \pi)$

$$\Rightarrow \theta = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \frac{\pi}{8}, \pi - \frac{\pi}{8}, \frac{3\pi}{8}, \pi - \frac{3\pi}{8}$$

$$\Rightarrow \theta = \frac{\pi}{8}, \frac{\pi}{3}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{2\pi}{3}, \frac{7\pi}{8}$$

7. Find the general solution of the equation, $2 + \tan x \cdot \cot \frac{x}{2} + \cot x \cdot \tan \frac{x}{2} = 0$.

समीकरण $2 + \tan x \cdot \cot \frac{x}{2} + \cot x \cdot \tan \frac{x}{2} = 0$ का व्यापक हल बताइए।

Ans. $x = 2n\pi \pm \frac{2\pi}{3}, n \in \mathbb{I}$

Sol. $2 + \tan x \cdot \cot \frac{x}{2} + \cot x \cdot \tan \frac{x}{2} = 0$

$$\Rightarrow 2 + \frac{\sin x}{\cos x} \cdot \frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} + \frac{\cos x}{\sin x} \cdot \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} = 0$$

$$\Rightarrow 2 + \frac{2\cos^2 \frac{x}{2}}{\cos x} + \frac{\cos x}{2\cos^2 \frac{x}{2}} = 0 \quad \left\{ \begin{array}{l} \because \cos x \neq 0 \\ \& 1 + \cos x \neq 0 \end{array} \right\}$$

$$\Rightarrow 2 + \frac{1 + \cos x}{\cos x} + \frac{\cos x}{1 + \cos x} = 0$$

$$\Rightarrow 2(\cos x + \cos^2 x) + 1 + \cos^2 x + 2\cos x + \cos^2 x = 0$$

$$\Rightarrow 4\cos^2 x + 4\cos x + 1 = 0$$

$$\Rightarrow (2\cos x + 1)^2 = 0$$

$$\Rightarrow \cos x = -\frac{1}{2}$$

$$\Rightarrow x = 2n\pi \pm \frac{2\pi}{3}; n \in \mathbb{I}$$

8. Sum of all the solutions in $[0, 4\pi]$ of the equation $\tan x + \cot x + 1 = \cos\left(x + \frac{\pi}{4}\right)$ is $k\pi$ then the value of k is.

अन्तराल $[0, 4\pi]$ में समीकरण $\tan x + \cot x + 1 = \cos\left(x + \frac{\pi}{4}\right)$ के सभी हलों का योग $k\pi$ है, तो k का मान ज्ञात कीजिये।

Ans. 3.5

Sol. Given $x \in [0, 4\pi]$

$$\tan x + \cot x + 1 = \cos\left(x + \frac{\pi}{4}\right)$$

$$\therefore \tan x + \cot x \in (-\infty, -2] \cup [2, \infty)$$

$$\Rightarrow \tan x + \cot x + 1 \in (-\infty, -1] \cup [3, \infty)$$

$$\& \cos\left(x + \frac{\pi}{4}\right) \in [-1, 1]$$

This is only possible only when

$$\cos\left(x + \frac{\pi}{4}\right) = -1 \quad \&$$

$$\tan x + \cot x + 1 = -1$$

$$\Rightarrow x + \frac{\pi}{4} = (2n+1)\pi \quad \&$$

$$\tan x + \cot x = -2$$

$$\Rightarrow x = (2n+1)\pi - \frac{\pi}{4} \quad \&$$

$$\tan^2 x + 2\tan x + 1 = 0$$

$\&$

$$(\tan x + 1)^2 = 0$$

$$\therefore x \in [0, 4\pi]$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{11\pi}{4} \quad \&$$

$$\tan x = -1$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{11\pi}{4} \quad \&$$

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}, \frac{11\pi}{4}$$

$$\Rightarrow x = \frac{3\pi}{4}, \frac{11\pi}{4}$$

$$\Rightarrow \text{Sum of solutions} = \frac{3\pi}{4} + \frac{11\pi}{4} = \frac{14\pi}{4} = \frac{7\pi}{2}$$

$$\Rightarrow k = \frac{7}{2} = 3.5$$

SOLVING TRIGONOMETRIC EQUATION TYPE 5 (MAXIMUM AND MINIMUM BASED)

9. Find the range of y such that the equation, $y + \cos x = \sin x$ has a real solution. For $y = 1$, find x such that $0 < x < 2\pi$.

y का परिसर ज्ञात कीजिए जिसके लिये समीकरण $y + \cos x = \sin x$ का कोई वास्तविक हल हो। $y = 1$ के लिये x का $(0 < x < 2\pi)$ में मान ज्ञात कीजिए।

Ans. $-\sqrt{2} \leq y \leq \sqrt{2} ; \frac{\pi}{2}, \pi$

Sol. $y + \cos x = \sin x$

$$\Rightarrow y = \sin x - \cos x$$

$$\Rightarrow \text{Range of } y \text{ is } [-\sqrt{2}, \sqrt{2}]$$

If $y = 1$ then $1 + \cos x = \sin x$

$$\Rightarrow \sin x - \cos x = 1$$

$$\Rightarrow \frac{\sin x}{\sqrt{2}} - \frac{\cos x}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} = \cos\left(\frac{3\pi}{4}\right)$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi + \frac{3\pi}{4} \quad \text{or} \quad x + \frac{\pi}{4} = 2n\pi - \frac{3\pi}{4}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{2}; n \in I \quad \text{or} \quad x = 2n\pi - \pi; n \in I$$

$$\therefore x \in (0, 2\pi)$$

$$\Rightarrow x = \frac{\pi}{2} \quad \text{or} \quad x = \pi$$

$$\Rightarrow x = \frac{\pi}{2}, \pi$$

10. Let $f(x) = \sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$ for some real number k . Determine :

- all real numbers k for which $f(x)$ is constant for all values of x .
- all real numbers k for which there exists a real number ' c ' such that $f(c) = 0$.
- If $k = -0.7$, determine all solutions to the equation $f(x) = 0$.

माना $f(x) = \sin^6 x + \cos^6 x + k(\sin^4 x + \cos^4 x)$, जहाँ k कोई वास्तविक संख्या है, तो :

- k के वह सभी वास्तविक मान ज्ञात कीजिये जिनके लिये $f(x)$, x के सभी मानों के लिये, अचर है।
- k के वह सभी वास्तविक मान ज्ञात कीजिये जिनके लिये एक वास्तविक संख्या ' c ' इस प्रकार विद्यमान है कि $f(c) = 0$ हो।

(c) यदि $k = -0.7$ है, तो समीकरण $f(x) = 0$ के सभी हल ज्ञात कीजिये।

Ans. (a) $-\frac{3}{2}$; (b) $k \in \left[-1, -\frac{1}{2}\right]$; (c) $x = \frac{n\pi}{2} \pm \frac{\pi}{6}$

Sol. $f(x) = \sin^6 x + \cos^6 x + k(9\sin^4 x + \cos^4 x)$
 $\Rightarrow f(x) = 1 - 3\sin^2 x \cos^2 x + k(1 - 2\sin^2 x \cos^2 x)$
 $\Rightarrow f(x) = (k+1) - \sin^2 x \cos^2 x (2k+3)$

(a) Given all real numbers k for which $f(x)$ is constant for all values of x

$$\Rightarrow 2k+3=0$$

$$\Rightarrow k = -\frac{3}{2}$$

(b) Given all real numbers k for which there exists a real number c such that $f(c) = 0$

$$\Rightarrow (k+1) - (2k+3) \sin^2 c \cos^2 c = f(c) = 0$$

$$\Rightarrow \frac{\sin^2(2c)}{4} = \frac{k+1}{2k+3}$$

$$\because \sin^2(2c) \in [0,1]$$

$$\Rightarrow 0 \leq \frac{4(k+1)}{2k+3} \leq 1$$

$$0 \leq \frac{4(k+1)}{2k+3} \quad \& \quad \frac{4k+4-2k-3}{2k+3} \leq 0$$

$$\Rightarrow \frac{4k+4}{2k+3} \geq 0 \quad \& \quad \frac{2k+1}{2k+3} \leq 0$$

$$\Rightarrow k \in \left(-\infty, -\frac{3}{2}\right) \cup [-1, \infty) \quad \& \quad k \in \left(-\frac{3}{2}, -\frac{1}{2}\right]$$

$$\Rightarrow k \in \left[-1, -\frac{1}{2}\right]$$

(c) If $k = -0.7 = \frac{-7}{10}$

then $f(x) = \left(\frac{-7}{10} + 1\right) - \sin^2 x \cos^2 x \left(\frac{-14}{10} + 3\right) = 0$

$$\Rightarrow \frac{3}{10} - \frac{16}{10} \sin^2 x \cos^2 x = 0$$

$$\Rightarrow \sin^2 2x = \frac{3}{4} = \left(\frac{\sqrt{3}}{2}\right)^2 = \sin^2 \frac{\pi}{3}$$

$$\Rightarrow 2x = n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x = \frac{n\pi}{2} \pm \frac{\pi}{6}; n \in I$$

SOLVING TRIGONOMETRIC EQUATION TYPE 6 (SQUARING BASED)

11. Solve the equation for x , $5^{\frac{1}{2}} + 5^{\frac{1}{2} + \log_5(\sin x)} = 15^{\frac{1}{2} + \log_{15} \cos x}$.

समीकरण $5^{\frac{1}{2}} + 5^{\frac{1}{2} + \log_5(\sin x)} = 15^{\frac{1}{2} + \log_{15} \cos x}$ को x के लिये हल कीजिए।

Ans. $x = 2n\pi + \frac{\pi}{6}, n \in I$

Sol. $5^{\frac{1}{2}} + 5^{\frac{1}{2} + \log_5(\sin x)} = 15^{\frac{1}{2} + \log_{15} \cos x}$

$$\Rightarrow 5^{\frac{1}{2}} + 5^{\frac{1}{2}} \cdot \sin x = 15^{\frac{1}{2}} \cdot \cos x$$

$$\Rightarrow 1 + \sin x = \sqrt{3} \cos x$$

$$\Rightarrow \sqrt{3} \cos x - \sin x = 1$$

$$\Rightarrow \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \frac{1}{2}$$

$$\Rightarrow \cos\left(x + \frac{\pi}{6}\right) = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\Rightarrow x + \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{6} \quad \text{or} \quad x = 2n\pi - \frac{\pi}{2}$$

$$\therefore \sin x > 0 \text{ \& \> } \cos x > 0$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{6}; n \in I$$

12. Solve for x , the equation $\sqrt{13 - 18 \tan x} = 6 \tan x - 3$, where $-2\pi < x < 2\pi$.

अन्तराल $-2\pi < x < 2\pi$ में समीकरण $\sqrt{13 - 18 \tan x} = 6 \tan x - 3$ के लिये x के मान निकालिए।

Ans. $\alpha - 2\pi$; $\alpha - \pi$, α , $\alpha + \pi$, where $\tan \alpha = \frac{2}{3}$

Sol. $\sqrt{13 - 18 \tan x} = 6 \tan x - 3; -2\pi < x < 2\pi$

square both sides; we get

$$\Rightarrow 13 - 18 \tan x = 36 \tan^2 x + 9 - 36 \tan x$$

$$\Rightarrow 36 \tan^2 x - 18 \tan x - 4 = 0$$

$$\Rightarrow 18 \tan^2 x - 9 \tan x - 2 = 0$$

$$\Rightarrow (3 \tan x - 2)(6 \tan x + 1) = 0$$

$$\Rightarrow \tan x = \frac{2}{3} \quad \text{or} \quad \tan x = \frac{-1}{6}$$

$$\left\{ \begin{array}{l} \text{For } \sqrt{13 - 18 \tan x} \quad 13 - 18 \tan x \geq 0 \\ \Rightarrow \tan x \leq \frac{13}{18} \\ \tan x \geq \frac{1}{2} \quad \text{as } 6 \tan x - 3 \geq 0 \end{array} \right\}$$

$$\Rightarrow \tan x = \frac{2}{3}$$

$$\Rightarrow x = n\pi + \alpha; \quad \alpha = \tan^{-1}\left(\frac{2}{3}\right)$$

$$\Rightarrow x = \alpha, \quad \pi + \alpha, \quad -\pi + \alpha, \quad -2\pi + \alpha$$

13. Determine the smallest positive value of x which satisfy the equation, $\sqrt{1 + \sin 2x} - \sqrt{2} \cos 3x = 0$.

समीकरण $\sqrt{1 + \sin 2x} - \sqrt{2} \cos 3x = 0$ को संतुष्ट करने वाले x की न्यूनतम धनात्मक मान बताइए।

Ans. $x = \pi/16$

Sol. $\sqrt{1 + \sin 2x} = \sqrt{2} \cos 3x$

If $\cos 3x \geq 0$

then

$$1 + \sin 2x = 2 \cos^2 3x$$

$$\Rightarrow \sin 2x = 2 \cos^2 3x - 1$$

$$\Rightarrow \sin 2x = \cos 6x$$

$$\Rightarrow \cos 6x = \cos\left(\frac{\pi}{2} - 2x\right)$$

$$\Rightarrow 6x = 2n\pi \pm \left(\frac{\pi}{2} - 2x\right)$$

$$\Rightarrow 8x = 2n\pi + \frac{\pi}{2} \quad \text{or} \quad 4x = 2n\pi - \frac{\pi}{2}; n \in \mathbb{I}$$

$$\Rightarrow x = \frac{n\pi}{4} + \frac{\pi}{16} \quad \text{or} \quad x = \frac{n\pi}{2} - \frac{\pi}{8}; n \in I$$

$\therefore$ Smallest positive value of x is $\frac{\pi}{16}$.

14. $2 \sin \left(3x + \frac{\pi}{4} \right) = \sqrt{1 + 8 \sin 2x \cdot \cos^2 2x}$.

Ans. $x = 2n\pi + \frac{\pi}{12}$ or $2n\pi + \frac{17\pi}{12}; n \in I$

Sol. $2 \sin \left(3x + \frac{\pi}{4} \right) = \sqrt{1 + 8 \sin 2x \cos^2 2x}$

$$\Rightarrow 2 \sin \left(3x + \frac{\pi}{4} \right) = \sqrt{1 + 4 \sin 4x \cos 2x}$$

$$\Rightarrow 4 \sin^2 \left(3x + \frac{\pi}{4} \right) = 1 + 4 \sin 4x \cos 2x. \text{ If } \sin \left(3x + \frac{\pi}{4} \right) \geq 0$$

$$\Rightarrow 2 - 2 \cos \left(6x + \frac{\pi}{2} \right) = 1 + 2 \sin 6x + 2 \sin 2x$$

$$\Rightarrow 2 + 2 \sin 6x = 1 + 2 \sin 6x + 2 \sin 2x$$

$$\Rightarrow \sin 2x = \frac{1}{2} \Rightarrow x = \frac{n\pi}{2} + (-1)^n \frac{\pi}{12}, n \in I$$

$$\Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$$

$$\therefore \sin \left(3x + \frac{\pi}{4} \right) \geq 0 \text{ so } n = \frac{\pi}{12}, \frac{17\pi}{12} \text{ satisfies}$$

$$\therefore \text{ general solution is } x = 2n\pi + \frac{\pi}{12}, 2n\pi + \frac{17\pi}{12}$$

15. Find the general solution of the trigonometric equation $3^{\left(\frac{1}{2} + \log_3 (\cos x + \sin x) \right)} - 2^{\log_2 (\cos x - \sin x)} = \sqrt{2}$.

त्रिकोणमितीय समीकरण $3^{\left(\frac{1}{2} + \log_3 (\cos x + \sin x) \right)} - 2^{\log_2 (\cos x - \sin x)} = \sqrt{2}$ का व्यापक हल ज्ञात कीजिए।

Ans. $x = 2n\pi + \frac{\pi}{12}$

Sol. $3^{\left(\frac{1}{2} + \log_3(\cos x + \sin x)\right)} - 2^{\log_2(\cos x - \sin x)} = \sqrt{2}$

$$\because \sin x + \cos x > 0 \quad \& \quad \cos x - \sin x > 0$$

$$\Rightarrow \sin x + \cos x > 0 \quad \& \quad \sin x - \cos x < 0$$

$$\text{In } (0, 2\pi)$$

$$x \in \left(\frac{7\pi}{4}, \frac{\pi}{4}\right)$$

Now

$$\sqrt{3}(\sin x + \cos x) - (\cos x - \sin x) = \sqrt{2}$$

$$\Rightarrow \frac{\sqrt{3}-1}{2\sqrt{2}} \cos x + \frac{\sqrt{3}+1}{2\sqrt{2}} \sin x = \frac{1}{2}$$

$$\Rightarrow \cos\left(x - \frac{5\pi}{12}\right) = \frac{1}{2} = \cos \frac{\pi}{3}$$

$$\Rightarrow x = \left(2n\pi \pm \frac{\pi}{3}\right) + \frac{5\pi}{12}$$

$$\because \sin x + \cos x > 0 \quad \& \quad \sin x - \cos x < 0$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{12}; n \in \mathbb{I}$$

16. If sum of all the solutions of the equation $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}} \cos 2x$

is equal to $\frac{k\pi}{6}$, then find the value of k.

यदि समीकरण $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}} \cos 2x$ के सभी हलों का योग $\frac{k\pi}{6}$ है, तो k का मान ज्ञात कीजिये।

Ans. 18

Sol. $\log_{\frac{9x-x^2-14}{7}}(\sin 3x - \sin x) = \log_{\frac{9x-x^2-14}{7}} \cos 2x$

For log

$$\frac{9x-x^2-14}{7} > 0 \quad \& \quad \frac{9x-x^2-14}{7} \neq 1$$

$$\Rightarrow x^2 - 9x + 14 < 0 \quad \& \quad x^2 - 9x + 21 \neq 0$$

$$\Rightarrow (x-7)(x-2) < 0 \quad \& \quad x^2 - 9x + 21 \neq 0$$

$$\Rightarrow x \in (2, 7)$$

For log

$$\sin 3x - \sin x > 0 \quad \& \quad \cos 2x > 0$$

$$\Rightarrow 2 \cos 2x \sin x > 0 \quad \& \quad \cos 2x > 0$$

$$\Rightarrow \sin x > 0 \quad \& \quad \cos 2x > 0$$

Now

$$\sin 3x - \sin x = \cos 2x$$

$$\Rightarrow 3 \sin x - 4 \sin^3 x - \sin x = 1 - 2\sin^2 x$$

$$\Rightarrow 4\sin^3 x - 2\sin^2 x - 2\sin x + 1 = 0$$

$$\Rightarrow (2\sin^2 x - 1)(2\sin x - 1) = 0$$

$$\Rightarrow \sin^2 x = \frac{1}{2} \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$\Rightarrow \sin^2 x = \sin^2 \frac{\pi}{4} \quad \text{or} \quad \sin x = \frac{1}{2}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{4}; n \in I \quad \text{or} \quad x = n\pi + (-1)^n \frac{\pi}{6}$$

$$\Rightarrow x = \frac{5\pi}{6}, \frac{13\pi}{6}$$

$$\Rightarrow \text{sum} = \frac{5\pi}{6} + \frac{13\pi}{6} = \frac{18\pi}{6}$$

$$\Rightarrow k = 18$$

NUMBER OF SOLUTION IN AN INTERVAL

17. If the quadratic equation $x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$ has two integral roots, then sum of all possible values of θ in interval $(0, 2\pi)$ is $k\pi$. Find the value of k .

यदि द्विघातीय समीकरण $x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0$ के दोनों मूल पूर्णांक हैं तो अन्तराल $(0, 2\pi)$ में θ के सारे सम्भावित मानों का योग $k\pi$ होगा तो k का मान ज्ञात करें।

Ans. 4

Sol. Given

Quadratic Equation

$$x^2 + (2 - \tan \theta)x - (1 + \tan \theta) = 0 \text{ has two integral roots.}$$

$$\text{Then } 2 - \tan \theta \in \mathbb{Q}$$

$$\& 1 + \tan \theta \in \mathbb{Q}$$

$$\& D \equiv (2 - \tan \theta)^2 + 4(1 + \tan \theta) \text{ must be perfect square.}$$

$$\Rightarrow D \equiv 4 + \tan^2 \theta + 4$$

$$\Rightarrow D \equiv \tan^2 \theta + 8; \text{ is perfect square}$$

$$\text{if } \tan^2 \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}, \dots$$

Given $\theta \in (0, 2\pi)$

$\Rightarrow$ Sum of all possible values of θ is

$$\frac{\pi}{4} + \frac{5\pi}{4} + \frac{3\pi}{4} + \frac{7\pi}{4} = \frac{16\pi}{4} = 4\pi$$

$$\Rightarrow k = 4$$

TRIGONOMETRIC INEQUALITIES

18. Solve the inequality $\sin 2x > \sqrt{2} \sin^2 x + (2 - \sqrt{2}) \cos^2 x$.

असमिका $\sin 2x > \sqrt{2} \sin^2 x + (2 - \sqrt{2}) \cos^2 x$ को हल करें।

Ans. $n\pi + \frac{\pi}{8} < x < n\pi + \frac{\pi}{4}$

Sol. $\sin 2x > \sqrt{2} \sin^2 x + (2 - \sqrt{2}) \cos^2 x$

$$\Rightarrow 2 \sin x \cos x > \sqrt{2} \sin^2 x + 2 \cos^2 x - \sqrt{2} \cos^2 x$$

$$\Rightarrow \frac{2 \sin x \cos x}{\cos^2 x} > \sqrt{2} \tan^2 x + 2 - \sqrt{2}$$

$$\Rightarrow \sqrt{2} \tan^2 x - 2 \tan x + 2 - \sqrt{2} < 0$$

$$\sqrt{2} (\tan^2 x - 1) - 2 (\tan x - 1) < 0$$

$$\Rightarrow (\tan x - 1) \{ \sqrt{2} (\tan x + 1) - 2 \} < 0$$

$$\Rightarrow (\tan x - 1) (\tan x + 1 - \sqrt{2}) < 0$$

$$\Rightarrow \tan x \in (\sqrt{2} - 1, 1)$$

$$\Rightarrow x \in \left(\frac{\pi}{8}, \frac{\pi}{4} \right)$$

$$\Rightarrow \text{General solution is } x \in \left(n\pi + \frac{\pi}{8}, n\pi + \frac{\pi}{4} \right)$$

SYSTEM OF TRIGONOMETRIC EQUATION

19. For $0 < x, y < \pi$, find the number of ordered pairs (x, y) satisfying the system of equations

$$\cot^2(x - y) - (1 + \sqrt{3}) \cot(x - y) + \sqrt{3} = 0 \text{ and } \cos y = \frac{\sqrt{3}}{2}.$$

यदि $0 < x, y < \pi$ हो, तो समीकरणों $\cot^2(x-y) - (1+\sqrt{3})\cot(x-y) + \sqrt{3} = 0$ तथा $\cos y = \frac{\sqrt{3}}{2}$ के निकाय को संतुष्ट

करने वाले क्रमित युग्मों (x, y) की संख्या ज्ञात कीजिये।

Ans. 2

Sol. Given $0 < x, y < \pi$

$$\cot^2(x-y) - (1+\sqrt{3})\cot(x-y) + \sqrt{3} = 0$$

$$\& \cos y = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \cot^2(x-y) - \cot(x-y) - \sqrt{3}\cot(x-y) + \sqrt{3} = 0$$

$$\Rightarrow (\cot(x-y) - 1)(\cot(x-y) - \sqrt{3}) = 0$$

$$\Rightarrow \cot(x-y) = 1 \quad \text{or} \quad \cot(x-y) = \sqrt{3}$$

$$\Rightarrow \tan(x-y) = 1 \quad \text{or} \quad \tan(x-y) = \frac{1}{\sqrt{3}}$$

$$\Rightarrow x-y = n\pi + \frac{\pi}{4} \quad \text{or} \quad x-y = n\pi + \frac{\pi}{6}$$

$$\because \cos y = \frac{\sqrt{3}}{2} = \cos\left(\frac{\pi}{6}\right)$$

$$\Rightarrow y = 2n\pi \pm \frac{\pi}{6}$$

$$\because 0 < x, y < \pi$$

$$\Rightarrow y = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\text{If } y = \frac{\pi}{6} \text{ then}$$

$$\Rightarrow x = n\pi + \frac{\pi}{4} + \frac{\pi}{6} \quad \text{or} \quad x = n\pi + \frac{\pi}{6} + \frac{\pi}{6}$$

$$\Rightarrow x = \frac{5\pi}{12} \quad \text{or} \quad x = \frac{\pi}{3}$$

$$\text{If } y = \frac{5\pi}{6} \text{ then}$$

$$x = n\pi + \frac{\pi}{4} + \frac{5\pi}{6} \quad \text{or} \quad x = n\pi + \pi$$

$$\therefore x \in (0, \pi)$$

$$\Rightarrow \text{There is no } x \text{ for } y = \frac{5\pi}{6}$$

$$\therefore \text{ Only possible pairs are } \left(\frac{5\pi}{12}, \frac{\pi}{6}\right), \left(\frac{\pi}{3}, \frac{\pi}{6}\right)$$

$$\Rightarrow \text{No. of ordered pairs are 2.}$$

20. If the number of solutions satisfying the equation $(\sin \theta - 1)(2 \sin \theta - 1)(3 \sin \theta - 1) \dots (n \sin \theta - 1) = 0$ (where $n \in \mathbb{N}$) in $[0, \pi]$ is 9, then find the number of ordered pairs (x, y) satisfying the equation

$$3 + \operatorname{cosec}^2 x + 2^{\sin^2 y} = n, \text{ where } 0 \leq x, y \leq 4\pi.$$

यदि समीकरण $(\sin \theta - 1)(2 \sin \theta - 1)(3 \sin \theta - 1) \dots (n \sin \theta - 1) = 0$ (जहाँ $n \in \mathbb{N}$) को संतुष्ट करने वाले हलों की संख्या

अन्तराल $[0, \pi]$ में 9 है, तो समीकरण $3 + \operatorname{cosec}^2 x + 2^{\sin^2 y} = n$, जहाँ $0 \leq x, y \leq 4\pi$, को संतुष्ट करने वाले क्रमित युग्मों (x, y) की संख्या ज्ञात कीजिए।

Ans. 20

Sol. Given numbers of solutions satisfying the equation $(\sin \theta - 1)(2 \sin \theta - 1) \dots (n \sin \theta - 1) = 0$ in $[0, \pi]$ is 9.

$\Rightarrow \sin \theta = 1$	or	$\sin \theta = \frac{1}{2}$	or	$\sin \theta = \frac{1}{3}$
$\downarrow$		$\downarrow$		$\downarrow$
one solution		two solution		two solution
 $\text{or } \sin \theta = \frac{1}{4}$	 or	 $\sin \theta = \frac{1}{5}$		
$\downarrow$		$\downarrow$		
two solution		two solution		

$$\Rightarrow n = 5$$

$$3 + \operatorname{cosec}^2 x + 2^{\sin^2 y} = n = 5; 0 \leq x, y \leq 4\pi$$

$$\Rightarrow \operatorname{cosec}^2 x + 2^{\sin^2 y} = 2$$

$$\therefore \operatorname{cosec}^2 x \geq 1 \quad \& \quad 2^{\sin^2 y} \geq 1$$

$\therefore$ This is possible only when

$$\operatorname{cosec}^2 x = 1 \quad \& \quad 2^{\sin^2 y} = 1$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{2} \quad \& \quad y = n\pi$$

$$\Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2} \quad \& \quad y = 0, \pi, 2\pi, 3\pi, 4\pi$$

$\therefore$ Number of ordered pairs (x, y) is $4 \times 5 = 20$.

21. Solve x and y : $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$, $5 \cdot 16^{\sin x} - 2 \cdot 3^{\frac{1}{\cos y}} = 2$

x तथा y ज्ञात कीजिए : $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$, $5 \cdot 16^{\sin x} - 2 \cdot 3^{\frac{1}{\cos y}} = 2$

Ans. $x = n\pi + (-1)^n \frac{\pi}{6}$ and $y = 2m\pi \pm \frac{\pi}{3}$, where $m, n \in I$

Sol. $4^{\sin x} + 3^{\frac{1}{\cos y}} = 11$

$$5 \cdot 16^{\sin x} - 2 \cdot 3^{\frac{1}{\cos y}} = 2$$

Let $4^{\sin x} = a$, $3^{\frac{1}{\cos y}} = b$

$$\Rightarrow a + b = 11 \quad \& \quad 5 \cdot a^2 - 2b = 2$$

$$\Rightarrow 5a^2 - 2(11 - a) - 2 = 0$$

$$\Rightarrow 5a^2 + 2a - 24 = 0$$

$$\Rightarrow (5a + 12)(a - 2) = 0$$

$$\Rightarrow a = 2 \quad \text{or} \quad a = \underbrace{\frac{-12}{5}}_{\text{Not possible}}$$

$$\Rightarrow 4^{\sin x} = 2 \quad \& \quad 3^{\sec y} = 9 = 3^2$$

$$\Rightarrow \sin x = \frac{1}{2} \quad \& \quad \sec y = 2$$

$$\Rightarrow \sin x = \frac{1}{2} \quad \& \quad y = 2m\pi \pm \frac{\pi}{3}$$

$$\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}; n \in I$$

$$\& y = 2m\pi \pm \frac{\pi}{3}; m \in I$$

22. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$ and $\tan \theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$, is. **[JEE-2010]**

अन्तराल $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में θ के मानों की संख्या जिसके लिये $\theta \neq \frac{n\pi}{5}$ जहाँ $n = 0, \pm 1, \pm 2$ तथा

$\tan \theta = \cot 5\theta$ साथ ही $\sin 2\theta = \cos 4\theta$ है।

[JEE-2010]

Ans. 3

Sol. Given $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ & $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$

$$\sin 2\theta = \cos 4\theta$$

$$\sin 2\theta = 1 - 2\sin^2 2\theta$$

$$2\sin^2 2\theta + \sin 2\theta - 1 = 0$$

$$\Rightarrow \sin 2\theta = -1 \text{ or } \sin 2\theta = \frac{1}{2}$$

$$\therefore \theta = -\frac{\pi}{4}, \frac{\pi}{12}, \frac{5\pi}{12}$$

Now $\tan \theta = \cot 5\theta$ (Given)

All three obtained values of θ satisfy the given equation.

Hence, number of values of θ are 3.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 23 to 24

Let $f(x) = \sin^3 x + 8 \cos^6 x - 3$, $g(x) = 2 + 3\sin 2x \cos x$ and $h(x) = \sin x + 3\cos 2x \sin x$.

On the basis of above information, answer the following questions :

माना $f(x) = \sin^3 x + 8 \cos^6 x - 3$, $g(x) = 2 + 3\sin 2x \cos x$ तथा $h(x) = \sin x + 3\cos 2x \sin x$.

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उत्तर दीजिए।

23. If $g(x) + h(x) = 0$, then number of solutions in the interval $x \in [0, 2\pi]$ is :

यदि $g(x) + h(x) = 0$, तो अन्तराल $x \in [0, 2\pi]$ में हलों की संख्या कितनी होगी :

(A) 1

(B) 3

(C) 5

(D) 6

Ans. C

Sol. $g(x) + h(x) = 0$

$$\Rightarrow 2 + 3(\sin 2x \cos x + \cos 2x \sin x) + \sin x = 0$$

$$\Rightarrow 2 + 3 \sin 3x + \sin x = 0$$

$$\Rightarrow 9 \sin x - 12 \sin^3 x + \sin x + 2 = 0$$

$$\Rightarrow 12 \sin^3 x - 10 \sin x - 2 = 0$$

$$\Rightarrow 6\sin^3 x - 5\sin x - 1 = 0$$

$$\Rightarrow (\sin x - 1)(6 \sin^2 x + 6 \sin x + 1) = 0$$

$$\Rightarrow \sin x = 1 \quad \text{or} \quad \sin x = \frac{-3 - \sqrt{3}}{6} \quad \text{or} \quad \sin x = \frac{-3 + \sqrt{3}}{6}$$

$$\text{In } 0 \leq x \leq 2\pi$$

$$\sin x = 1$$

$\rightarrow$

only one solution

$$\sin x = \frac{-3 - \sqrt{3}}{6} \rightarrow \text{two solutions}$$

$$\sin x = \frac{-3 + \sqrt{3}}{6} \rightarrow \text{two solutions}$$

$\therefore$ No. of solution is 5.

24. If $f(x) + g(x) = 0$, then number of solutions in the interval $x \in \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$ is :

यदि $f(x) + g(x) = 0$, तो अन्तराल $x \in \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$ में हलों की संख्या कितनी होगी :

- (A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. $f(x) + g(x) = 0$; $x \in \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$

$$\Rightarrow \sin^3 x + 8 \cos^6 x - 3 + 2 + 3 \sin 2x \cos x = 0$$

$$\Rightarrow (\sin x)^3 + (2 \cos^2 x)^3 + (-1)^3 = 3 \sin x (2 \cos^2 x) (-1)$$

$$\Rightarrow \sin x + 2 \cos^2 x - 1 = 0 \text{ or } \underbrace{\sin x = -1 = 2 \cos^2 x}_{\text{Not possible}} \quad \left\{ \begin{array}{l} \because a^3 + b^3 + c^3 = 3abc \\ \text{if } a = b = c \text{ or } a + b + c = 0 \end{array} \right\}$$

$$\Rightarrow \sin x + 2(1 - \sin^2 x) - 1 = 0$$

$$\Rightarrow 2 \sin^2 x - \sin x - 1 = 0$$

$$\Rightarrow (\sin x - 1)(2 \sin x + 1) = 0$$

$$\Rightarrow \sin x = 1 \text{ or } \sin x = \frac{-1}{2}$$

$$\text{In } \left[-\frac{\pi}{2}, \frac{5\pi}{2}\right]$$

Number of solutions is 5.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

25. If $(\sin 2x + \sqrt{3} \cos 2x)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ then the value of $\cos x$ can be

यदि $(\sin 2x + \sqrt{3} \cos 2x)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ हो, तो $\cos x$ का मान हो सकता है -

- (A) $\frac{\sqrt{6} + \sqrt{2}}{4}$ (B) $\frac{\sqrt{6} - \sqrt{2}}{4}$ (C) $-\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right)$ (D) $\frac{\sqrt{2} - \sqrt{6}}{4}$

Ans. BD

Sol. $(\sin 2x + \sqrt{3} \cos 2x)^2 = 5 + \cos\left(2x - \frac{\pi}{6}\right)$

$$\Rightarrow 4\left(\frac{\sin 2x}{2} + \frac{\sqrt{3}}{2} \cos 2x\right)^2 = 5 + \cos\left(2x - \frac{\pi}{6}\right)$$
$$\Rightarrow 4\cos^2\left(2x - \frac{\pi}{6}\right) - \cos\left(2x - \frac{\pi}{6}\right) = 5$$
$$\Rightarrow 4\cos^2\left(2x - \frac{\pi}{6}\right) - \cos\left(2x - \frac{\pi}{6}\right) - 5 = 0$$
$$\Rightarrow \left(4\cos\left(2x - \frac{\pi}{6}\right) - 5\right)\left(\cos\left(2x - \frac{\pi}{6}\right) + 1\right) = 0$$
$$\Rightarrow \underbrace{\cos\left(2x - \frac{\pi}{6}\right) = \frac{5}{4}}_{\text{Not possible as } \frac{5}{4} > 1.} \quad \text{or} \quad \cos\left(2x - \frac{\pi}{6}\right) = -1$$

$$\Rightarrow \cos\left(2x - \frac{\pi}{6}\right) = -1$$

$$\Rightarrow 2x - \frac{\pi}{6} = (2n-1)\pi$$

$$\Rightarrow x = (2n-1)\frac{\pi}{2} + \frac{\pi}{12}$$

$$\Rightarrow x = n\pi - \frac{5\pi}{12}; n \in \mathbb{I}$$

$$\Rightarrow x = -\frac{5\pi}{12}, \frac{7\pi}{12}, \frac{-17\pi}{12}, \dots$$

$$\cos\left(\frac{7\pi}{12}\right) = \cos\left(\pi - \frac{5\pi}{12}\right) = -\cos\frac{5\pi}{12} = \frac{\sqrt{2}-\sqrt{6}}{4}$$

$$\cos\left(-\pi - \frac{5\pi}{12}\right) = \cos\left(\pi + \frac{5\pi}{12}\right) = -\cos\left(\frac{5\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$$

26. If $|\cos x|^{\sin^2 x - \frac{3}{2} \sin x + \frac{1}{2}} = 1$. Then possible values of x are :

यदि $|\cos x|^{\sin^2 x - \frac{3}{2} \sin x + \frac{1}{2}} = 1$ तब x के सम्भव मान होंगे :

(A) $n\pi$ or $n\pi + (-1)^n \frac{\pi}{6}$, $n \in I$

(B) $n\pi$ or $2n\pi + \frac{\pi}{2}$ or $n\pi + (-1)^n \frac{\pi}{6}$, $n \in I$

(C) $n\pi + (-1)^n \frac{\pi}{6}$, $n \in I$

(D) $n\pi$, $n \in I$

Ans. ACD

Sol. $|\cos x|^{\sin^2 x - \frac{3}{2} \sin x + \frac{1}{2}} = 1$

Case I :- $\sin^2 x - \frac{3}{2} \sin x + \frac{1}{2} = 0$

$\Rightarrow 2 \sin^2 x - 3 \sin x + 1 = 0$

$\Rightarrow (2 \sin x - 1)(\sin x - 1) = 0$

$\Rightarrow \sin x = \frac{1}{2}$ or $\sin x = 1$

$\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}$ or $x = (4n+1)\frac{\pi}{2}$; $n \in I$

Case II :- $|\cos x| = 1$

$\Rightarrow x = n\pi$

27. If $\frac{\sin^2 2x + 4 \sin^4 x - 4 \sin^2 x \cos^2 x}{4 - \sin^2 2x - 4 \sin^2 x} = \frac{1}{9}$ and $0 < x < \pi$, then value of x is :

यदि $\frac{\sin^2 2x + 4 \sin^4 x - 4 \sin^2 x \cos^2 x}{4 - \sin^2 2x - 4 \sin^2 x} = \frac{1}{9}$ तथा $0 < x < \pi$, तब x के सम्भव मान होंगे :

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{6}$

(C) $\frac{2\pi}{3}$

(D) $\frac{5\pi}{6}$

Ans. BD

Sol. $\frac{\sin^2 2x + 4 \sin^4 x - 4 \sin^2 x \cos^2 x}{4 - \sin^2 2x - 4 \sin^2 x} = \frac{1}{9}$; $0 < x < \pi$

$\Rightarrow \frac{\sin^2 2x + 4 \sin^4 x - \sin^2 2x}{4(1 - \sin^2 x \cos^2 x - \sin^2 x)} = \frac{1}{9}$

$\Rightarrow 9 \sin^4 x = 1 - \sin^2 x + \sin^4 x - \sin^2 x$

$\Rightarrow 8 \sin^4 x = 1 - 2 \sin^2 x$

$\Rightarrow 8 \sin^4 x + 2 \sin^2 x - 1 = 0$

$\Rightarrow 8 \sin^4 x + 4 \sin^2 x - 2 \sin^2 x - 1 = 0$

$\Rightarrow (4 \sin^2 x - 1)(2 \sin^2 x + 1) = 0$

$$\Rightarrow \sin^2 x = \frac{1}{4} \quad \text{or} \quad \underbrace{\sin^2 x = \frac{-1}{2}}_{\text{Not possible}}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{6}$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6} \quad \{\because 0 < x < \pi\}$$

28. $5 \sin x - 12 \cos x = -13 \sin 3x$, if $\phi = \sin^{-1}\left(\frac{12}{13}\right)$, then which of the following options are correct :

यदि $\phi = \sin^{-1}\left(\frac{12}{13}\right)$, तथा $5 \sin x - 12 \cos x = -13 \sin 3x$ हो तो निम्न में से कौनसा विकल्प सही है :

(A) $\sin\left(2x - \frac{\phi}{2}\right) = 0$ (B) $\sin\left(x + \frac{\phi}{2}\right) = 0$ (C) $\cos\left(x + \frac{\phi}{2}\right) = 0$ (D) $\cos\left(2x + \frac{\phi}{2}\right) = 0$

Ans. AC

Sol. $5 \sin x - 12 \cos x = -13 \sin 3x$ & $\phi = \sin^{-1}\left(\frac{12}{13}\right)$

$$\Rightarrow \frac{5}{13} \sin x - \frac{12}{13} \cos x = -\sin 3x$$

$$\Rightarrow \sin x \cos \phi - \cos x \sin \phi = -\sin 3x$$

$$\Rightarrow \sin(x - \phi) + \sin 3x = 0$$

$$\Rightarrow 2 \sin\left(\frac{4x - \phi}{2}\right) \cos\left(\frac{2x + \phi}{2}\right) = 0$$

$$\Rightarrow \sin\left(2x - \frac{\phi}{2}\right) = 0 \quad \text{or} \quad \cos\left(x + \frac{\phi}{2}\right) = 0$$

29. If $\sin 5\theta = a \sin^5 \theta + b \sin^3 \theta + c \sin \theta + d$, $\forall \theta \in \mathbb{R}$, then :

यदि $\sin 5\theta = a \sin^5 \theta + b \sin^3 \theta + c \sin \theta + d$, $\forall \theta \in \mathbb{R}$ तब :

(A) $a + b + c + d = 1$ (B) $a + b + c = 1$ (C) $5a + 4b = 0$ (D) $b + 4c = 0$

Ans. ABCD

Sol. Given $\sin 5\theta = a \sin^5 \theta + b \sin^3 \theta + c \sin \theta + d$ $\forall \theta \in \mathbb{R}$

$$\Rightarrow \sin 5\theta = \sin(3\theta + 2\theta)$$

$$= \sin 3\theta \cos 2\theta + \cos 3\theta \sin 2\theta$$

$$= 2 \sin \theta \cos \theta (4 \cos^3 \theta - 3 \cos \theta) + (1 - 2 \sin^2 \theta) (3 \sin \theta - 4 \sin^3 \theta)$$

$$= 8 \sin \theta \cos^4 \theta - 6 \sin \theta \cos^2 \theta + 3 \sin \theta - 4 \sin^3 \theta - 6 \sin^3 \theta + 8 \sin^5 \theta$$

$$\begin{aligned}
&= 8 \sin \theta + 8 \sin^5 \theta - 16 \sin^3 \theta - 3 \sin \theta - 4 \sin^3 \theta + 8 \sin^5 \theta \\
&= 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta \\
\Rightarrow a &= 16, b = -20, c = 5, d = 0 \\
\Rightarrow a + b + c + d &= 1 \\
a + b + c &= 1 \\
5a + 4b &= 80 - 80 = 0 \\
&\& b + 4c = 0
\end{aligned}$$

MATCH THE COLUMN TYPE QUESTION(S)

30. Match the following column :

Column I

- (A) Solution of $|\tan \theta| + |\cot \theta| \leq 2$ is
- (B) Solution of $(\sqrt{2} \cos \theta + 1)^2 + (\tan \theta - 1)^2 = 0$ is
- (C) Solution of $\sin \left(\frac{11}{10} \cos \theta \right) = \cos \left(\frac{11}{10} \sin \theta \right)$ is
- (D) Solution of $\sin x = 2^{x-1} + 2^{1-x}$ is

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Column I

- (A) $|\tan \theta| + |\cot \theta| \leq 2$ का हल होगा
- (B) $(\sqrt{2} \cos \theta + 1)^2 + (\tan \theta - 1)^2 = 0$ का हल होगा
- (C) $\sin \left(\frac{11}{10} \cos \theta \right) = \cos \left(\frac{11}{10} \sin \theta \right)$ का हल होगा
- (D) $\sin x = 2^{x-1} + 2^{1-x}$ का हल होगा

Column II

- (P) $2n\pi + \frac{5\pi}{4}, n \in I$
- (Q) Null set
- (R) $(2n+1)\frac{\pi}{4}, n \in I$
- (S) $n\pi + (-1)^n \frac{\pi}{6}, n \in I$

Column II

- (P) $2n\pi + \frac{5\pi}{4}, n \in I$
- (Q) रिक्त समुच्चय
- (R) $(2n+1)\frac{\pi}{4}, n \in I$
- (S) $n\pi + (-1)^n \frac{\pi}{6}, n \in I$

Ans. (A) R, (B) P, (C) Q, (D) Q

Sol. (A) $|\tan \theta| + |\cot \theta| \leq 2$

Let $\tan \theta = t$

$$\Rightarrow |t| + \frac{1}{|t|} \leq 2$$

$$\left\{ \because t + \frac{1}{t} \geq 2 \& t + \frac{1}{t} \leq -2 \right.$$

$$\Rightarrow |t| + \frac{1}{|t|} = 2$$

$$\Rightarrow (|t| - 1)^2 = 0$$

$$\Rightarrow \tan \theta = 1 \quad \text{or}$$

$$\tan \theta = -1$$

$$\Rightarrow \theta = n\pi + \frac{\pi}{4} \quad \text{or}$$

$$\theta = n\pi - \frac{\pi}{4}$$

For $n = 0$

$$\theta = \frac{\pi}{4} \quad \text{or}$$

$$\theta = -\frac{\pi}{4}$$

For $n = 1$

$$\theta = \frac{5\pi}{4} \quad \text{or}$$

$$\theta = \frac{3\pi}{4}$$

$$\Rightarrow \theta = (2n+1)\frac{\pi}{4}; n \in I$$

$$(B) (\sqrt{2} \cos \theta + 1)^2 + (\tan \theta - 1)^2 = 0$$

$$\Rightarrow \tan \theta = 1 \quad \&$$

$$\sqrt{2} \cos \theta + 1 = 0$$

$$\Rightarrow \tan \theta = 1 \quad \&$$

$$\cos \theta = \frac{-1}{\sqrt{2}}$$

$$\Rightarrow \theta = n\pi + \frac{\pi}{4} \quad \&$$

$$\theta = 2n\pi - \frac{3\pi}{4}; n \in I$$

$$\Rightarrow \theta = 2n\pi + \frac{5\pi}{4}; n \in I$$

$$(C) \sin\left(\frac{11}{10}\cos\theta\right) = \cos\left(\frac{11}{10}\sin\theta\right)$$

$$\Rightarrow \frac{11}{10}\cos\theta = \frac{\pi}{2} - \frac{11}{10}\sin\theta$$

$$\Rightarrow \frac{11}{10}(\cos\theta + \sin\theta) = \frac{22}{14}$$

$$\Rightarrow \cos\theta + \sin\theta = \frac{10}{7}$$

$$\therefore \cos\theta + \sin\theta \in [-\sqrt{2}, \sqrt{2}]$$

$$\text{but } \frac{10}{7} > \sqrt{2}$$

$\therefore$ There is no solution.

$$(D) \sin x = 2^{x-1} + 2^{1-x}$$

$$\Rightarrow \underbrace{\sin x}_{\downarrow [-1,1]} = \underbrace{\frac{2^x}{2} + \frac{2}{2^x}}_{\in [2, \infty)}$$

$\Rightarrow$ No solution

31. Match the following column :

Column I (Equation)

(A) $\sin x + \cos x = 1$

(B) $\sin x + \cos x = -1$

(C) $|\sin x| + |\cos x| = 1$

(D) $|\sin x + \cos x| = 1$

Column II (General Solution)

(P) $2n\pi, n \in I$

(Q) $(2n+1)\pi, n \in I$

(R) $(4n+1)\frac{\pi}{2}, n \in I$

(S) $(4n-1)\frac{\pi}{2}, n \in I$

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Column I (समीकरण)

(A) $\sin x + \cos x = 1$

(B) $\sin x + \cos x = -1$

(C) $|\sin x| + |\cos x| = 1$

(D) $|\sin x + \cos x| = 1$

Column II (व्यापक हल)

(P) $2n\pi, n \in I$

(Q) $(2n+1)\pi, n \in I$

(R) $(4n+1)\frac{\pi}{2}, n \in I$

(S) $(4n-1)\frac{\pi}{2}, n \in I$

Ans. (A) PR, (B) QS, (C) PQRS, (D) PQRS

Sol. (A) $\sin x + \cos x = 1$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \cos \frac{\pi}{4}$$

$$\Rightarrow x = \left(2n\pi \pm \frac{\pi}{4}\right) + \frac{\pi}{4}; n \in I$$

$$\Rightarrow x = 2n\pi \quad \text{or} \quad x = 2n\pi + \frac{\pi}{2}; n \in I$$

(B) $\sin x + \cos x = -1$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \cos \frac{3\pi}{4}$$

$$\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}$$

$$\Rightarrow x = (2n+1)\pi \quad \text{or} \quad x = (4n-1)\frac{\pi}{2}; n \in I$$

$$(C) |\sin x| + |\cos x| = 1$$

In first quad.

$$\sin x + \cos x = 1$$

$$\Rightarrow \cos\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \cos \frac{\pi}{4}$$

$$\Rightarrow x = 2n\pi \quad \text{or} \quad (4n+1)\frac{\pi}{2}; n \in I$$

In second quad.

$$\sin x - \cos x = 1$$

$$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = \cos \frac{3\pi}{4}$$

$$\Rightarrow x = (4n+1)\frac{\pi}{2} \quad \text{or} \quad x = (2n-1)\pi; n \in I$$

In third quad.

$$\sin x + \cos x = -1$$

$$x = (2n+1)\pi \quad \text{or} \quad x = (4n-1)\frac{\pi}{2}; n \in I$$

In fourth quad.

$$-\sin x + \cos x = 1$$

$$\Rightarrow \cos x - \sin x = 1$$

$$\Rightarrow \cos\left(x + \frac{\pi}{4}\right) = \cos \frac{\pi}{4}$$

$$\Rightarrow x = 2n\pi; n \in I \quad \text{or} \quad x = (4n-1)\frac{\pi}{2}; n \in I$$

$$(D) |\sin x + \cos x| = 1$$

$$\Rightarrow \sin x + \cos x = 1 \quad \text{or} \quad \sin x + \cos x = -1$$

$$\Rightarrow x = 2n\pi \quad \text{or} \quad x = (2n+1)\pi \quad \text{or} \quad x = (4n+1)\frac{\pi}{2} \quad \text{or} \quad x = (4n-1)\frac{\pi}{2}$$

32. If α and β are the roots of the equation, $a\cos\theta + b\sin\theta = c$ then match the entries of column-I with the entries of column-II :

यदि α तथा β समीकरण $a \cos \theta + b \sin \theta = c$ के मूल हैं तो स्तम्भ का मिलान कीजिए :

Column-I

(A) $\sin \alpha + \sin \beta$

(B) $\sin \alpha \cdot \sin \beta$

(C) $\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}$

(D) $\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} =$

Column-II

(P) $\frac{2b}{a+c}$

(Q) $\frac{c-a}{c+a}$

(R) $\frac{2bc}{a^2+b^2}$

(S) $\frac{c^2-a^2}{a^2+b^2}$

Ans. (A) R; (B) S; (C) P; (D) Q

Sol. $a \cos \theta + b \sin \theta = c$

$$\Rightarrow a \cos \theta = c - b \sin \theta$$

square both sides

$$\Rightarrow a^2 \cos^2 \theta = c^2 + b^2 \sin^2 \theta - 2bc \sin \theta$$

$$\Rightarrow \underbrace{(a^2 + b^2) \sin^2 \theta - 2bc \sin \theta + (c^2 - a^2)}_{\text{Quadratic Equation have roots } \sin \alpha \text{ \& } \sin \beta} = 0$$

$$\Rightarrow \sin \alpha + \sin \beta = \frac{2bc}{a^2 + b^2}$$

$$\& \sin \alpha \cdot \sin \beta = \frac{c^2 - a^2}{a^2 + b^2}$$

Now $a \cos \theta + b \sin \theta = c$

$$\Rightarrow \frac{a \left(1 - \tan^2 \frac{\theta}{2} \right)}{1 + \tan^2 \frac{\theta}{2}} + \frac{2b \left(\tan \frac{\theta}{2} \right)}{1 + \tan^2 \frac{\theta}{2}} = c$$

Let $\tan \frac{\theta}{2} = t$

$$\Rightarrow \frac{a(1-t^2)}{(1+t^2)} + \frac{2bt}{(1+t^2)} = c$$

$$\Rightarrow a - at^2 + 2bt - c - ct^2 = 0$$

$$\Rightarrow (a+c)t^2 - 2bt + (c-a) = 0$$

$$\Rightarrow (a+c)\tan^2 \frac{\theta}{2} - 2b \tan \frac{\theta}{2} + c - a = 0$$

$\underbrace{\hspace{10em}}$
 Quadratic Equation have roots $\tan \frac{\alpha}{2}$ & $\tan \frac{\beta}{2}$

$$\Rightarrow \tan \frac{\alpha}{2} + \tan \frac{\beta}{2} = \frac{2b}{a+c}$$

$$\tan \frac{\alpha}{2} \cdot \tan \frac{\beta}{2} = \frac{c-a}{a+c}$$